MMIF: Interpretable Hyperspectral and Multispectral Image Fusion via Maximum Mutual Information

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***Abstract*— Fusion-based hyperspectral image (HSI) super-resolution (SR) aims to recover high-resolution HSI (hr-HSI) from its two degraded modalities, that is, low-resolution HSI (lr-HSI) and high-resolution multispectral image (hr-MSI). The resulting HSI-SR image can be explained from two viewpoints, an inverse problem solution or a spatial–spectral information fusion product. Recent methods focusing on the former point are limited by demands of accurate degradation parameters and precise prior assumptions. On the contrary, the other research line avoids these drawbacks and offers a more flexible design. However, recent methods implicitly handle information fusion. The interaction between lr-HSI and hr-MSI only lies in the extracted feature domain, not directly on the data. Moreover, the proposed modules in recent methods promote information fusion from the perspective of deep learning and neglect the specialty of the HSI domain, leading to weak interpretability and poor reliability in practice. Considering the essence of the HSI-SR problem and the inherent property of HSIs, in this article, we propose a maximum mutual information (MMI) strategy. At first, we model the HSI-SR problem in a new variance inference (VI) architecture. This new VI model simulates the physical process of HSI pairs imaging and provides convenience for the MMI strategy. Then, we insert the MMI strategy in the VI model. The MMI strategy promotes information fusion with quantitative constraints on the information interaction between lr-HSI and hr-MSI. Finally, we implement the VI architecture with a neural network. Benefitting from the MMI strategy, a simple network structure can achieve efficient fusion performance, which indicates that MMI frees DL-based HSI-SR methods from the complicated structure design. The experimental results on synthetic and real datasets demonstrate the superiority of our method to state-of-the-art methods in terms of effectiveness, generalization, and interpretability.**

***Index Terms*— Deep learning, hyperspectral image (HSI) super-resolution (SR), image fusion, maximum mutual information (MMI), variance inference (VI).**

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I. Introduction

DUE to high spectral resolution, hyperspectral images (HSIs) can provide intrinsic spectral properties of different materials. Consequently, HSI analysis has promoted a wide range of applications, such as tracking, object detection, and classification [1], [2], [3]. However, the physical limitations of image sensors cause a tradeoff between spectral and spatial resolution. Consequently, HSIs have much lower spatial resolution than multispectral images (MSIs), which impairs the advantage of HSIs in practical applications [4], [5]. Therefore, HSI super-resolution (SR) has recently attracted significant attention. Currently, many satellites are equipped with HSI and MSI imaging payloads. A reliable way to achieve HSI-SR is fusing a high-resolution MSI (hr-MSI, henceforth, if not otherwise specified, resolution refers to the spatial resolution) with the observed low-resolution HSI (lr-HSI). These two observations come from different sensors under the same scene. Compared with single-image SR (SISR), HSI-SR offers better restoration quality and broader application scenarios.

HSI-SR can be explained from two perspectives: building an inverse problem solver or realizing spatial–spectral information fusion.

As shown in Fig. 1, the observed lr-HSI and hr-MSI can be viewed as spatial degradation and spectral degradation of the desired hr-HSI. Following the physical degeneration processes of hr-HSI to lr-HSI and hr-MSI, we can build an inverse process to recover hr-HSI. In this viewpoint, HSI-SR is an inverse problem. Unfortunately, the necessity of degeneration parameters [i.e., point spread function (PSF) and spectral response function (SRF)] limits the practicability of this research line. These degeneration parameters are challenging to acquire in real cases. Meanwhile, HSI-SR is a serious ill-posed problem. Previous methods have to explore image priors to shrink the solution space [6], [7], [8], [9]. The handcraft priors may not constantly adapt hr-HSI flexibly in diverse testing scenes [10].

The spatial and spectral information of the to-be-recovered hr-HSI mainly lies in hr-MSI and lr-HSI, respectively. Thus, modeling HSI-SR as an information fusion problem is a reasonable approach. In this viewpoint, the specific degradation process is not required, which avoids the limitations in the research line of solving the inverse problem theoretically. From the implementation of information fusion, recent HSI-SR methods can be traditional optimization methods

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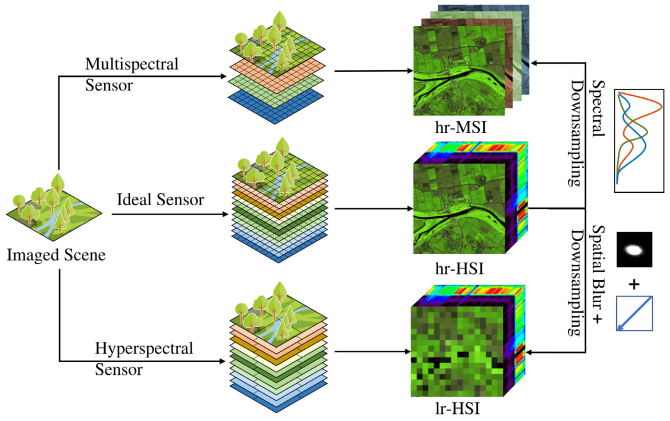


Fig. 1. Physical imaging process of images in HSI-SR. The spectral vector on a pixel is the combination of spectral vectors of all materials in the imaged area of this pixel, that is, the so-called spectral mixing phenomenon. Limited by the hardware factor, a tradeoff between spatial and spectral resolution exists in imaging. lr-HSI and hr-MSI can be viewed as spatial and spectral degradation of hr-HSI, respectively.

or deep-learning-based methods. Traditional methods process information fusion with a clear insight into the physical essence of HSIs. Coming from the spectral mixing phenomenon in HSI imaging, matrix factorization and subspace decomposition-based were utilized for information fusion by simulating the unmixing mechanism [11], [12]. Despite their clear interpretability, the weak ability of data representation limits their performance. It causes a demand for handcraft priors and precise degradation to constrain the final results [6], [7], [8], [9], [13], impairs the advantage of information fusion. Deep-learning-based methods attracted increasing attention in recent years. Inspired by the traditional methods, recent deep-learning-based methods realize information fusion through the unmixing mechanism with building coupled-autoencoders [14], [15], [16], [17]. The performance of these methods heavily depends on the consistency of features in two autoencoders. In other words, the divergence of two autoencoders' features leads to a significant degradation in the final performance. To overcome this problem, various modifications are proposed, such as cross-attention [15], alternative optimization [14], spatial–spectral variational inference [16], and KL-divergence losses [17]. However, these variants may not keep the original interpretation and do not eliminate the divergence completely.

The motivation behind our approach is mainly twofold. **First**, traditional methods implement information fusion with a clear physical meaning consistent with the real imaging process but limited by the weak representation of data. DL-based methods are excellent at learning data features but lack interpretability. These two methods are complementary. It is intuitive to explore a new approach that combines the advantages of these two research lines. **Second**, these two research lines are both fusing information indirectly. Specifically, these methods estimate the hr-HSI in two steps: information extraction and information fusion. The interaction path of lr-HSI and hr-MSI is built on the extracted features domain. Information loss is inevitable in both the information extraction and information fusion steps. It is necessary to find a new method that explicitly realizes the interaction between lr-HSI and hr-MSI and accomplishes information fusion in one shot.

In this article, we propose to combine the advantages of these two research lines and avoid their weakness. To integrate domain knowledge of HSI-SR in deep-learning methods, we first build a new variance inference (VI) architecture. The VI architecture simulates the physical generation in HSI-SR, which is easy to unfold to a neural network. Moreover, based on the VI architecture, it is convenient to calculate the image distribution in HSI-SR. Second, to achieve information extraction and fusion in one shot, we explore a maximum mutual information (MMI) strategy. The MMI strategy offers a direct information interaction path between hr-MSI and lr-HSI. In the end, we implement our model with a neural network, which holds one-to-one corresponding network modules and model components, leading to better interpretability. Extensive experiments on synthetic and real datasets demonstrate the superiority of our method to state-of-the-art methods in terms of effectiveness, generalization, and interpretability.

The remainder of this article is organized as below. In Section II, we briefly review existing work on HSI-SR from information fusion. The proposed MMI fusion (MMIF) method is detailed in Section III. In Section IV, we report the experimental results to demonstrate the effectiveness and efficiency of our method. Section V concludes this article.

II. Related Work

A. Information Fusion in Numerical Optimization-Based HSI-SR

Most numerical optimization-based approaches formulate information fusion through matrix factorization, tensor factorization, or subspace decomposition. In matrix factorization-based methods, lr-HSI and hr-MSI are decomposed to spectral signature matrices and abundance matrices, which hold spectral and spatial information of original images, respectively [11], [18]. Information fusion is realized by matrix multiplication on a spectral signature matrix of lr-HSI and an abundance matrix of hr-MSI. By modeling each image as a 3-D tensor, tensor factorization methods extract spectral and spatial information, where the HSI-SR problem was transformed to estimation of core tensor and dictionaries [7], [8], [9]. Similar to matrix factorization, subspace decomposition methods handle HSI-SR in matrix form [13]. These methods achieve information fusion in linear transforms, leading to limited data representation. Meanwhile, HSI-SR is an ill-posed problem. Poor representation means the methods cannot handle the individual features of images. As a consequence, the solution is far away from the true one. Thus, a handcraft prior can be added to confine the solution space, such as sparsity [6], [7], [8], low-rank [9], and vector total variation [12]. The reconstruction constrain can also impose a demand for accurate information on PSF and SRF.

B. Information Fusion in Deep-Learning-Based HSI-SR

Promoted by the powerful data representation capacity of neural networks, deep-learning-based methods achieve state-of-the-art performance in HSI-SR.

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Recent supervised deep-learning-based methods learn the information fusion model from training datasets and apply it to the testing images. Palsson et al. [19] focused on the 3-D nature of HSIs and conducted 3D-CNN for HSI-SR. Promoted by the deep unfolding technique, model-driven methods were proposed and achieved state-of-the-art performance [10], [20], [21], [22]. Through the existing unfolding model in HSI-SR, model-driven methods merge intrinsic prior and data representation in an entire interpretable framework. The superiority of supervised learning methods relies on two factors, sufficient training image pairs, and consistent training and testing scenarios. Limited datasets and diverse scenes degrade their effectiveness in real applications. Most recently, attention mechanisms and transformers have become a fashion in HSI-SR. The spatial–spectral correlation between lr-HSI and hr-HSI can be effectively extracted by attention mechanisms [23], [24], [25]. Moreover, the inherent nonlocal feature extraction of the transformer promotes global spatial–spectral correlation discovery [26], [27]. Although these methods have achieved good performance, their practicality is limited due to the high computational burden of transformers and attention mechanisms. Additionally, the complex network structures decrease interpretability.

Recent unsupervised deep-learning methods handle information fusion in two steps: information extraction and fusion. Coupled-autoencoder is a common structure in unsupervised methods, where features of lr-HSI and hr-MSI are extracted independently. The core of this method lies in the alignment of features. In [14], an alternative optimization strategy was proposed. In each training period, several epochs were used to optimize feature extraction, and the other epochs were used to align inner features. In [15], cross-attention was inserted into two autoencoders for information interaction. In [16], spatial–spectral independent module and spatial-dependent prior were built for feature extraction. These complicated modifications harm the interpretability of these methods. More seriously, the coupled autoencoder handles fusion in two steps and each step may loss information. In detail, overfitting in feature extraction of autoencoders and misalignment in feature distribution are two factors that cause loss of information.

III. Proposed Variance Architecture With the MMI Strategy

Following the physical generation of HSIs and MSIs, we build a new VI architecture. Then, we introduce the MMI strategy into the VI architecture for the first time. We present our method in three parts. In Section III-A, we provide the physical generation process in HSI-SR. In Section III-B, we establish the VI architecture. In Section III-C, we introduce the MMI strategy.

A. Physical Generation of Images in HSI-SR

The physical process of images is shown in Fig. 1. Let $\mathcal{Z} \in \mathbb{R}^{H\times W\times L}$ denote the to-be recovered hr-HSI with H, W, and L as its height, width, and band number, respectively. The observed image pair of hr-MSI and lr-HSI are denoted as $\mathcal{Y} \in \mathbb{R}^{H\times W\times l}$ and $\mathcal{X} \in \mathbb{R}^{h\times w\times L}$, respectively, where $h \ll H, w \ll W, l \ll L$.

1) In Terms of Image Imaging: Each spectral vector is a combination of reflected lights from different materials in the pixel. The generation of lr-HSI, hr-MSI, and hr-HSI follows a spectral mixing mechanism and can be formulated as

$$\begin{cases} \mathcal{Z} = \mathrm{Em}(\mathcal{A}) \\ \mathcal{X} = \mathrm{Em}(\tilde{\mathcal{A}}) \\ \mathcal{Y} = \tilde{\mathrm{Em}}(\mathcal{A}). \end{cases} \tag{1}$$

$\mathcal{A} \in \mathbb{R}^{H\times W\times K}$ is the so-called abundance, which represents the component of material categories in each pixel of $\mathcal{Z}$ and $\mathcal{Y}$. K represents the number of material categories. $\tilde{\mathcal{A}} \in \mathbb{R}^{h\times w\times K}$ is the abundance of the lr-HSI $\mathcal{X}$. $\tilde{\mathcal{A}}$ can be viewed as the spatial degradation of $\mathcal{A}$. Em() and $\tilde{\mathrm{Em}}$() are nonlinear functions that simulate the spectral mixing phenomenon which combines different materials' spectrum to corresponding hyperspectral and multispectral spectral curves, respectively. Different from previous methods with linear combination [11], [15], [18], we simulate the spectral mixing mechanism in a nonlinear process, which is more closer to real cases. The relationship between $\tilde{\mathcal{A}}$ and $\mathcal{A}$ depends on the correlation between the HSI sensor and the MSI sensor. We formulate it as

$$\tilde{\mathcal{A}} = \mathbf{B} \odot \mathcal{A} \downarrow_{\mathrm{spatial}} \tag{2}$$

where $\mathbf{B}$ is a blur kernel that reflects the relationship between spatial imaging characteristics of HSI and MSI sensors.

2) In Terms of Image Degradation: lr-HSI and hr-MSI are the spatial and spectral degradation of hr-HSI, respectively. We formulate these degradation processes as

$$\begin{aligned} \mathcal{X} &= D_{\mathrm{spa}}(\mathcal{Z}) + \mathcal{N}_x \\ \mathcal{Y} &= D_{\mathrm{spe}}(\mathcal{Z}) + \mathcal{N}_y. \end{aligned} \tag{3}$$

$D_{\mathrm{spa}}()$ and $D_{\mathrm{spe}}()$ represent the nonlinear spatial and spectral degradation processes of $\mathcal{Z}$, respectively. $\mathcal{N}_x$ and $\mathcal{N}_y$ represent the noise during degradation. Following the widely recognized degradation model in HSI-SR, we set the noise as Gaussian noise.

B. VI Architecture

In HSI-SR, we estimate the unknown hr-HSI based on the observed lr-HSI and hr-MSI. VI model fits HSI-SR well. With setting lr-HSI and hr-MSI as observed variables, and hr-HSI as latent variables, the inference part of the VI model can estimate the latent hr-HSI under the constraint of the physical spectral mixing mechanism. The generative part reconstructs lr-HSI and hr-MSI from the estimated hr-HSI following the physical degradation process.

We build the VI architecture at the pixel level. Let $\boldsymbol{x}_{i,j} \in \mathcal{R}^L$ denote the pixel in the ith row and the jth column of lr-HSI. We represent the corresponding blocks in hr-MSI and hr-HSI as $\boldsymbol{y}^b_{i,j} \in \mathcal{R}^{s\times s\times l}$ and $\boldsymbol{z}^b_{i,j} \in \mathcal{R}^{s\times s\times L}$, respectively. s is the scaled factor that measures the proportion of spatial resolution between hr-MSI and lr-HSI, that is, $s = H/h = W/w$. The graphical illustration of VI is shown in Fig. 2.

The solid lines indicate the generative models $p_{\boldsymbol{\theta}_x}(\boldsymbol{x}_{i,j}|\boldsymbol{z}^b_{i,j})$ and $p_{\boldsymbol{\theta}_y}(\boldsymbol{y}^b_{i,j}|\boldsymbol{z}^b_{i,j})$, corresponding to the degradation of hr-HSI to lr-HSI and hr-MSI, respectively. For notation convenience,

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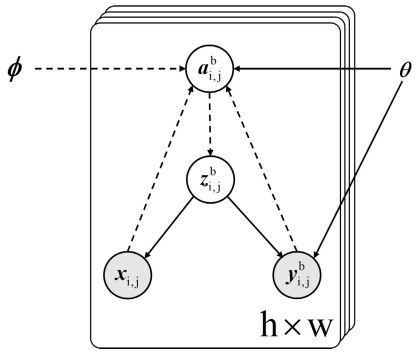


Fig. 2. Graphical illustration of the proposed VI architecture. Observed and latent variables are represented as circles with gray and white background, respectively. $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ represent the optimal parameters. The subscript of the box indicates that there are $h \times w$ data pairs for analysis.

we simply write $\boldsymbol{x}_{i,j}$, $\boldsymbol{y}_{i,j}^{b}$, and $\boldsymbol{z}_{i,j}^{b}$ as $\boldsymbol{x}$, $\boldsymbol{y}$, and $\boldsymbol{z}$, respectively. Based on (3), we formulate these physical degradation processes in distribution form

$$\begin{aligned} p_{\boldsymbol{\theta}_x}(\boldsymbol{x}|\boldsymbol{z}) &= \mathcal{N}\left(\boldsymbol{x}|D_{\text{spa}}(\boldsymbol{z};\boldsymbol{\theta}_x), \boldsymbol{\sigma}_x^2 \boldsymbol{I}_L\right) \\ p_{\boldsymbol{\theta}_y}\left(\boldsymbol{y}_{m,n}|\boldsymbol{z}_{m,n}\right) &= \mathcal{N}\left(\boldsymbol{y}|D_{\text{spe}}\left(\boldsymbol{z}_{m,n};\boldsymbol{\theta}_y\right), \boldsymbol{\sigma}_y^2 \boldsymbol{I}_l\right). \end{aligned} \tag{4}$$

The variance $\boldsymbol{\sigma}_x^2 \boldsymbol{I}_L$ and $\boldsymbol{\sigma}_y^2 \boldsymbol{I}_l$ comes from the noise in degradation, as shown in (3). m, n mean the mth row and the nth column pixel in the corresponding data block.

The dashed lines represent the inference model $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})$, which aims to estimate the latent hr-HSI through approximating the true posterior distribution $p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})$. Following the spectral mixing phenomenon in imaging, the inference model consists of two parts, $q_{\boldsymbol{\phi}}(\boldsymbol{a}|\boldsymbol{x}, \boldsymbol{y})$ and $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{a})$, corresponding to predicting the material categories of each pixel and mapping the material categories to spectral curves, respectively. Similar to [28], we force the approximated posterior distributions obeying normal distribution parameterized by $\boldsymbol{\phi}$

$$\begin{aligned} q_{\boldsymbol{\phi}_a}(\boldsymbol{a}|\boldsymbol{x}, \boldsymbol{y}) &= \mathcal{N}\left(\boldsymbol{a}|\text{Am}\left(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\phi}_a\right), \boldsymbol{\sigma}_a^2 \boldsymbol{I}_K\right) \\ q_{\boldsymbol{\phi}_e}(\boldsymbol{z}|\boldsymbol{a}) &= \mathcal{N}\left(\boldsymbol{z}|\text{Em}\left(\boldsymbol{a}; \boldsymbol{\phi}_e\right), \boldsymbol{\sigma}_z^2 \boldsymbol{I}_L\right) \end{aligned} \tag{5}$$

where $\text{Am}(\boldsymbol{x}, \boldsymbol{y}; \boldsymbol{\phi}_a)$ is a detector that predicts material components of each pixel, that is, the abundance. $\boldsymbol{a}$ is the abundance of the corresponding block. ϕ_a and ϕ_e correspond to the parameters of abundance prediction and endmember estimation, respectively.

For a precise approximation, we should minimize the distance between approximated and real posterior distribution as below

$$\min_{\boldsymbol{\phi}} D_{\text{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})||p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})). \tag{6}$$

We measure the distance through the commonly used KL divergence. However, the above objective function is intractable. On the one hand, the distribution of $p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})$ is unknown. On the other hand, the calculation of KL divergence presents a huge challenge to performing gradient descent on it. Fortunately, we can derive an equivalent tractable objective function from the likelihood of $\boldsymbol{x}, \boldsymbol{y}$

$$\log\ p(\boldsymbol{x}, \boldsymbol{y}) = D_{\text{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})||p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})) + \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \boldsymbol{x}, \boldsymbol{y}) \tag{7}$$

where $\boldsymbol{x}, \boldsymbol{y}$ are deterministic samples, leading to a fixed value of $\log\ p(\boldsymbol{x}, \boldsymbol{y})$. Thus, (6) can be transformed to maximize the second RHS (right-hand side) term of (7), the evidence lower bound (ELBO)

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\phi}; \boldsymbol{x}, \boldsymbol{y}) = E_{\boldsymbol{z}\sim q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x},\boldsymbol{y})}[\log\ p_{\boldsymbol{\theta}}(\boldsymbol{y}|\boldsymbol{z}) + \log\ p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})] \\ - D_{\text{KL}}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})||p_{\boldsymbol{\theta}}(\boldsymbol{z})) \end{aligned} \tag{8}$$

where $E_{p(x)}[f(x)]$ represents the expectation of $f(x)$ with respect to variate x with probability density $p(x)$. The first part of ELBO can be interpreted as a negative reconstruction error, and the second part prevents approximated posterior $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}, \boldsymbol{y})$ not too far away from the prior $p_{\boldsymbol{\theta}}(\boldsymbol{z})$. The precise prior distribution of $\boldsymbol{z}$ is unknown. However, we can utilize the spectral mixing prior. First, we insert the spectral mixing mechanism in the inference part, as shown in (5). Second, from the mixing phenomenon in imaging, the abundance $\boldsymbol{a}$ obeys the sum-to-one constraint (ASC). We utilize this prior knowledge as a prior constraint. Naturally, the object function of the VI architecture is

$$\mathcal{L}_{VI} = \gamma(||\boldsymbol{x} - \hat{\boldsymbol{x}}||_2 + ||\boldsymbol{y} - \hat{\boldsymbol{y}}||_2) + (1 - \gamma)||\boldsymbol{1}^T \cdot \boldsymbol{a} - 1||_1 \tag{9}$$

where $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ are reconstructed lr-HSI and hr-MSI in the generative part. γ is a parameter that balances the efforts of the reconstruction and spectral mixing prior.

C. MMI Strategy

hr-HSI contains the spectral information of lr-HSI and the spatial information of hr-MSI. The key factor of HSI-SR is to discover the spectral–spatial correlation between lr-HSI and hr-MSI. Naturally, we can grasp the relationship between lr-HSI and hr-MSI by maximizing the mutual information between them

$$\max_{\boldsymbol{\phi}} I(X, Y) \tag{10}$$

where $I(X, Y)$ is the mutual information between HSI space X and MSI space Y. It can be calculated as

$$I(X, Y) = -H(X|Y) + H(X) \tag{11}$$

where $H(X) = -E_x[\log p(\boldsymbol{x})]$ is the entropy of X, and $H(X|Y) = -E_{x,y}[\log p(\boldsymbol{x}|\boldsymbol{y})]$ is the conditional entropy of X depends on Y. $H(X)$ is the expectation of the information of samples $\boldsymbol{x}$ in X. $H(X|Y)$ measures the expectation information of $\boldsymbol{x}$ when $\boldsymbol{y}$ is known. Thus, $I(X, Y)$ quantifies the increase of information about the lr-HSI when hr-MSI is known. Compared with hr-MSI, lr-HSI possesses more spectral information. With maximizing the mutual information, the spectral information of lr-HSI is learned by the mapping function. The flow diagram of the MMI strategy is shown in Fig. 3.

In (11), we assume that the entropy of X is a fixed value. Thus, maximizing mutual information is equivalent to minimizing conditional entropy $H(X|Y)$. However, it is difficult to minimize $H(X|Y)$ because the exact calculation

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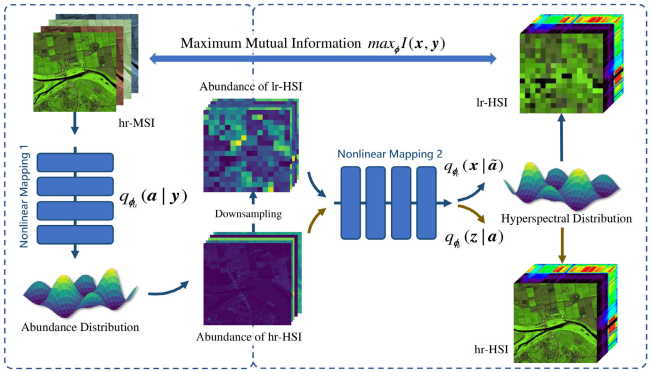


Fig. 3. Illustration of the proposed MMI strategy. Under the constraint of MMI between hr-MSI and lr-HSI, the mapping function $p_{\phi}(\boldsymbol{x}|\boldsymbol{y})$ aims to recover the lr-HSI from hr-HSI. The spectral information of lr-HSI is learned by the parameter of $p_{\phi}(\boldsymbol{x}|\boldsymbol{y})$, $\boldsymbol{\phi}$. We can recover the desired hr-HSI through $p_{\phi}(\boldsymbol{z}|\boldsymbol{y})$ with the same parameter $\boldsymbol{\phi}$.

is intractable. We transform to minimize the upper bound of $H(X|Y)$

$$\begin{aligned} H(X|Y) &= -E_{\boldsymbol{x},\boldsymbol{y}\sim p(\boldsymbol{x},\boldsymbol{y})}[\log p(\boldsymbol{x}|\boldsymbol{y})] \\ &= -E_{\boldsymbol{x},\boldsymbol{y}\sim p(\boldsymbol{x},\boldsymbol{y})}[\log q(\boldsymbol{x}|\boldsymbol{y})] \\ &\quad - E_{\boldsymbol{y}\sim p(\boldsymbol{y})}[D_{\mathrm{KL}}(p(\boldsymbol{x}|\boldsymbol{y})||q(\boldsymbol{x}|\boldsymbol{y}))] \\ &\leq -E_{\boldsymbol{x},\boldsymbol{y}\sim p(\boldsymbol{x},\boldsymbol{y})}[\log q(\boldsymbol{x}|\boldsymbol{y})]. \end{aligned} \tag{12}$$

This method is known as the *variational information maximization* [29]. A mapping between lr-HSI and hr-MSI already exists in the proposed VI architecture. Based on (5), we can build a mapping function between lr-HSI and hr-MSI

$$q_{\boldsymbol{\phi}}(\boldsymbol{x}|\boldsymbol{y}) = \int q_{\boldsymbol{\phi}_e}(\boldsymbol{x}|\tilde{\boldsymbol{a}}) q_{\boldsymbol{\phi}_a}(\boldsymbol{a}|\boldsymbol{y}) d\boldsymbol{a} \tag{13}$$

where $\tilde{\boldsymbol{a}}$ is the spatial degradation of $\boldsymbol{a}$. Inserting (5) and (2)–(13), we can find the conditional probability of lr-HSI given hr-MSI as follows:

$$\begin{aligned} &q_{\boldsymbol{\phi}}(\boldsymbol{x}|\boldsymbol{y}) \\ &= \mathcal{N}\big(\boldsymbol{x}|\mathrm{Em}\big(\mathbf{B} \odot (Am(\boldsymbol{y};\boldsymbol{\phi}) + \boldsymbol{\epsilon}_a) \downarrow_{\mathrm{spatial}};\boldsymbol{\phi}\big), \sigma_x^2 \boldsymbol{I}_L\big) \end{aligned} \tag{14}$$

where $\boldsymbol{\epsilon}_a$ samples from $\mathcal{N}(0, \boldsymbol{\sigma}_{\boldsymbol{a}} \mathbf{I}_K)$, which is derived from (5). To reduce the calculation complexity, we approximate $-E_{\boldsymbol{x},\boldsymbol{y}}[\log q(\boldsymbol{x}|\boldsymbol{y})]$, $\boldsymbol{x}$, $\boldsymbol{y}$ through the Monte Carlo method

$$-E_{\boldsymbol{x},\boldsymbol{y}\sim p(\boldsymbol{x},\boldsymbol{y})}\big[\log q(\boldsymbol{x}|\boldsymbol{y})\big] \approx -\frac{1}{MN}\sum_{i=1}^{H}\sum_{j=1}^{W} \log q_{\boldsymbol{\phi}}\big(\boldsymbol{x}_{i,j}|\boldsymbol{y}_{i,j}^{b}\big). \tag{15}$$

Naturally, we can derivate the loss function of the MMI strategy as

$$\mathcal{L}_{\mathrm{MMI}} = \frac{1}{NM}\sum_{i=1}^{N}\sum_{i=1}^{N}\left\|\boldsymbol{x}_{i,j} - \boldsymbol{x}_{i,j}^{\boldsymbol{\phi}}\right\| \tag{16}$$

where $\boldsymbol{x}_{i,j}^{\boldsymbol{\phi}}$ is the pixel in the ith row and the jth column of the estimated lr-HSI.

IV. Model Implementation

We first introduce the detailed calculation process in the VI architecture. Then, we will give the implementations of each operation in the flowchart.

A. Detailed Calculation Process

As shown in Fig. 4, there are three parts in the proposed VI architecture, the inference model, the MMI strategy, and the generative model.

In the inference model, we first map the posterior distribution of variable $\boldsymbol{a}$ from $\boldsymbol{y}$, which means inferring the abundance of observed hr-MSI and hr-HSI

$$\boldsymbol{a} = \mathrm{Am}_{\boldsymbol{\mu}}(\boldsymbol{y}) + \mathrm{Am}_{\boldsymbol{\sigma}}(\boldsymbol{y}) \cdot \boldsymbol{\epsilon}_{\boldsymbol{a}} \tag{17}$$

where $\boldsymbol{\epsilon}_{\boldsymbol{a}}$ is a sample from $\mathcal{N}(\mathbf{0}, \boldsymbol{I}_{\boldsymbol{a}})$. We utilize the reparameterization trick proposed in [28] to sample $\boldsymbol{a}$ from the estimated distribution. Subsequently, the distribution of $\boldsymbol{z}$ is inferred from $\boldsymbol{a}$

$$\boldsymbol{z} = \mathrm{Em}_{\boldsymbol{\mu}}(\boldsymbol{a}) + \mathrm{Em}_{\boldsymbol{\sigma}}(\boldsymbol{a}) \cdot \boldsymbol{\epsilon}_{\boldsymbol{z}}. \tag{18}$$

In this step, the spectral vector of each pixel is estimated based on the categories of materials contained in it. $\mathrm{Em}_{\boldsymbol{\mu}}$ and $\mathrm{Em}_{\boldsymbol{\sigma}}$ describe the spectral of different materials.

The MMI strategy aims to approximate the distribution of $\boldsymbol{x}$ from $\boldsymbol{y}$. Corresponding to (13), we first calculate $\tilde{\boldsymbol{a}}$ from $\boldsymbol{a}$ based on (2), and we set the blur kernel B as an optimizable parameter. The mapping from $\tilde{\boldsymbol{a}}$ to $\boldsymbol{x}$ holds the same parameter of $\boldsymbol{a}$ to $\boldsymbol{z}$

$$\boldsymbol{x}^{\boldsymbol{\phi}} = \mathrm{Em}_{\boldsymbol{\mu}}(\tilde{\boldsymbol{a}}) + \mathrm{Em}_{\boldsymbol{\sigma}}(\tilde{\boldsymbol{a}}) \cdot \boldsymbol{\epsilon}_{\boldsymbol{x}}. \tag{19}$$

Here, $\boldsymbol{x}^{\boldsymbol{\phi}}$ is a sample from the approximated distribution and is to be used in (16) to measure the distance between an approximated distribution and real distribution.

In the generative model, we reconstruct lr-HSI and hr-MSI from the estimated hr-HSI as

$$\begin{aligned} \hat{\boldsymbol{x}} &= D_{\mathrm{spa}}(\boldsymbol{z}) \\ \hat{\boldsymbol{y}} &= D_{\mathrm{spe}}(\boldsymbol{z}) \end{aligned} \tag{20}$$

where D_{spa} and D_{spe} simulate the degradation of hr-HSI to lr-HSI and hr-MSI, respectively.

B. Neural Network Implementation

There are seven operations in the VI architecture, $\mathrm{Am}_{\boldsymbol{\mu}}()$, $\mathrm{Am}_{\boldsymbol{\sigma}}()$, $\mathrm{Em}_{\boldsymbol{\mu}}()$, $\mathrm{Em}_{\boldsymbol{\sigma}}()$, $D_{\mathrm{spa}}()$, $D_{\mathrm{spe}}()$, and the blur kernel B. As shown in Fig. 4, we implement each operation with a corresponding neural network for a powerful data representation and convenient optimization by gradient descent. The detailed structure is shown in Table I.

$\mathrm{Am}_{\boldsymbol{\mu}}()$ aims to map hr-MSI to its corresponding abundance, which demands higher capacity of data representation. We implement $\mathrm{Am}_{\boldsymbol{\mu}}()$ through a convolution network with five layers. For nonlinear mapping, each layer is followed by a nonlinear activation function, leaky ReLu. To avoid overfitting in training, we skip the input hr-MSI with feature maps of each layer. In consideration of the abundance nonnegative constraint (ANC), we utilize a ReLu function on the sampled abundance. $\mathrm{Am}_{\boldsymbol{\sigma}}()$ introduces random factors in reference and is implemented by a lighter convolution network with two layers.

We implement $\mathrm{Em}_{\boldsymbol{\mu}}()$ and $\mathrm{Em}_{\boldsymbol{\sigma}}()$ by two nonlinear convolution networks. Compared to linear operations in [15],

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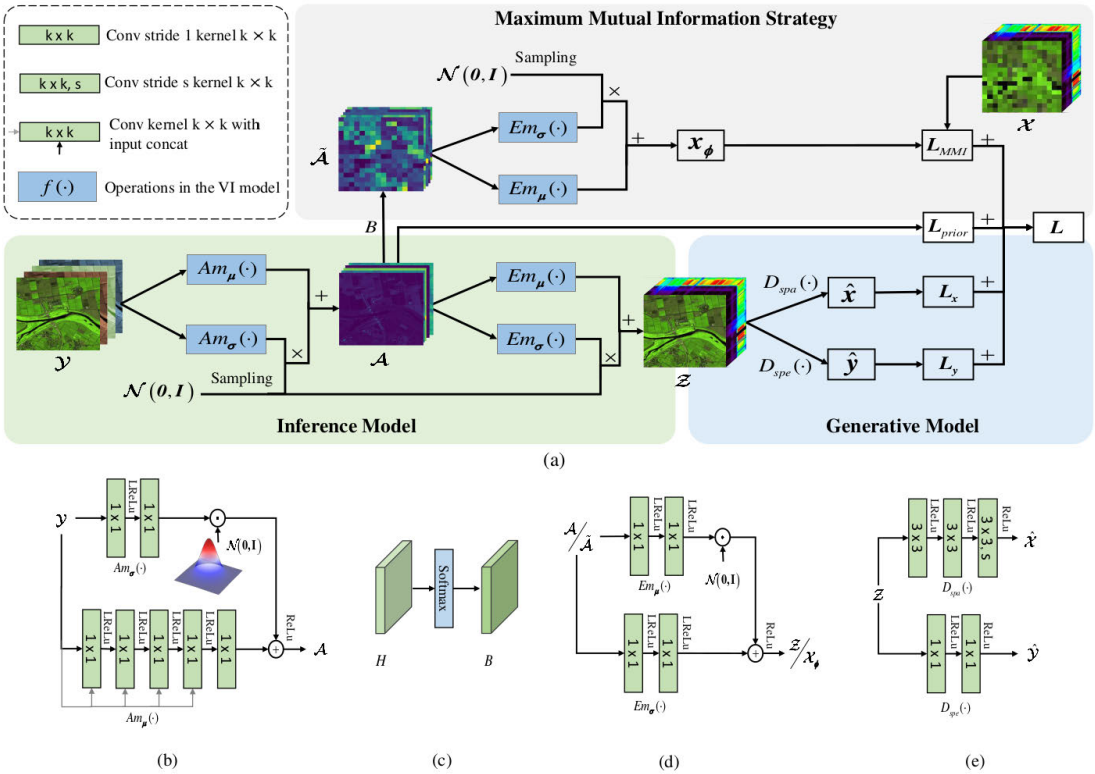


Fig. 4. Architecture of the proposed MMI fusion method (MMIF) for HSI-SR. "Conv stride s kernel k * k" represents a convolution layer with kernel size k*k and stride S. The architectures of (a) overall calculation flowchart, (b) abundance inference, (c) blur kernel estimation, (d) HSIs inference, and (e) generation model.

nonlinear spectral mapping is more applicable in real scenes with nonlinear spectral mixing. Similar to the implementation of $\mathrm{Am}_{\mu}()$, we apply leaky ReLU activation behind each layer. After the ending output of these two networks, a ReLU activation is leveraged for a nonnegative recovery of the hr-HSI.

D_{spa} and D_{spe} are spatial downsampling and spectral downsampling, respectively. We realize D_{spa} through a multilayer convolution network with the total stride equal to the scale factor. D_{spe} is simulated by a fully connected network with the channels being reduced to the number of bands in hr-MSI.

The summation of all elements in the blur kernel B should be equal to 1. We utilize a Softmax layer to realize this constraint

$$B = \mathrm{Softmax}(H) \tag{21}$$

where H is a trainable parameter with the same shape as B.

C. Network Training

The overall network is trained by minimizing the following loss function:

$$\mathcal{L} = \alpha \mathcal{L}_{\mathrm{VI}} + \beta \mathcal{L}_{\mathrm{MMI}} \tag{22}$$

where α and β are constants to tradeoff the effects of different terms. We set $\alpha = 0.1, \beta = 0.5$ in the experiments. In the VI loss $\mathcal{L}_{\mathrm{VI}}$, we set $\gamma = 0.5$ to balance the reconstruction constraint and prior constraint. In the MMI loss $\mathcal{L}_{\mathrm{MMI}}$, we set $N = 1$ samples per pixel under batch training, which is similar to [28].

V. Experiments

First, we demonstrate each mechanism in MMIF with experiments on synthetic datasets quantitatively. Then, we evaluate the performance of MMIF on synthetic and real datasets by comparing it to the existing state-of-the-art (SOTA) methods. Finally, we analyze the effort of several parameter settings in MMIF and dataset synthesis.

A. Experimental Setup

1) Implementation Details: We implement our network with the PyTorch 1.15 framework. Each 4×4 block in lr-HSI and its corresponding block in hr-MSI are set as a batch. We train the network in an unsupervised way with a batch size of 32. The parameters of the network are initialized by Kaiming Initialization [30]. We use the Adam [31] optimizer in the training stage with a learning rate 10^{-3} and a step decay of 0.5 at every 200 epochs.

2) Comparison Methods: For a comprehensive comparison, we select seven SOTA fusion based HSI-SR methods, including matrix factorization-based approaches, CNMF [11], Bayesian-based approaches, FUSE [13] and Hysure [12], tensor factorization-based approaches, CSTF [8] and LTTR [9], model-driven supervised learning approach, MHFNet [10], unsupervised learning approaches, CUCaNet [15], and UAL [32].

3) Evaluation Metrics: We adopt four picture quality indices (PQIs) for quantitative evaluation, including the peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM)

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TABLE I
DETAILED NETWORK ARCHITECTURE OF THE FUSION MODULE

dataset		CAVE	Chikusei	WV 2
module	operation	kernel size		
Am_μ	CONV+ LReLU	29×3	60×4	13×3
	Concat	HR-MSI		
	CONV+ LReLU	21×32	124×64	5×16
	Concat	HR-MSI		
	CONV+ LReLU	15×24	92×128	6×8
	Concat	HR-MSI		
	CONV+ LReLU	9×18	60×96	4×9
	Concat	HR-MSI		
	CONV+ LReLU	8×12	16×64	8×7
Am_σ	CONV+ LReLU	16×3	8×4	8×3
	CONV+ LReLU	8×16	16×8	4×8
Em_μ	CONV+ LReLU	16×8	64×16	6×4
	CONV+ ReLU	31×16	128×64	8×6
Em_σ	CONV+ LReLU	16×8	64×16	6×4
	CONV+ ReLU	31×16	128×64	8×6

TABLE II
ABLATION STUDY ON DIFFERENT MODULES IN MMIF (BEST RESULTS ARE HIGHLIGHTED IN BOLD)

Module			Ori CAVE		Syn CAVE	
Unmixing	MMI	NLG	PSNR	SAM	PSNR	SAM
✓		✓	15.61	63.90	17.46	62.45
	✓	✓	37.34	6.01	38.51	6.13
✓	✓		36.91	6.03	38.36	6.19
	✓		35.80	6.81	37.37	6.67
✓	✓	✓	**38.04**	**5.99**	**38.79**	**5.98**

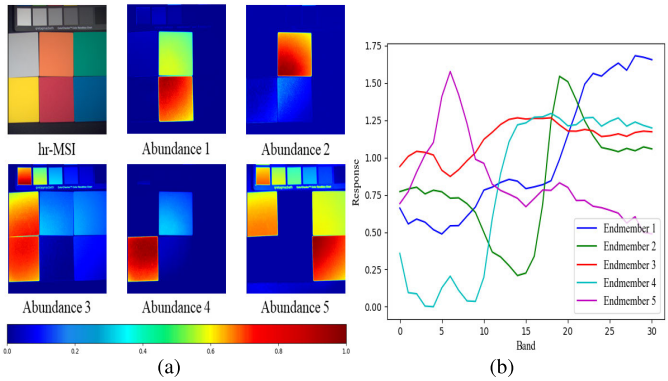


Fig. 5. Insights of unmixing mechanism in Am(·) and Em(·). (a) Input hr-MSI and its abundances produced by Am(·). (b) Corresponding endmembers (basic spectral vectors) produced by Em(·).

[33], relative dimensionless global error in synthesis (ERGAS) [34], and structure similarity (SSIM) [35]. SAM evaluates the spectral similarity between the estimated and reference image by calculating the average spectral angle of corresponding pixels. PSNR, ERGAS, and SSIM reflect the spatial similarity between the estimated and reference images specific to global confidence, organoleptic consistency, and structural consistency. The smaller ERGAS and SAM are, and the larger PSNR and SSIM are, the better the fusion result is.

B. Model Validation

We validate MMIF and its components on the CAVE dataset [36]. The CAVE dataset contains 32 image pairs imaging from 32 different indoor scenes. Each HSI has 512×512 pixels and 31 spectral bands measured in the wavelength ranging from 400 to 700 nm. The corresponding MSI is an RGB image rendered under a neutral daylight illuminant. We set the scale factor as 32 and simulated the lr-HSI under Gaussian blur kernel with the same size as the scale factor and 0.5 value deviation. Besides the given MSIs, we generated another MSI employing SRFs of a Nikon D700 camera. Then, we have two versions of the CAVE dataset, that is, the original CAVE dataset (ori CAVE) and the synthetic CAVE dataset (syn CAVE).

1) Ablation Study: We evaluated the effect of three components in MMIF, unmixing mechanism, MMI strategy, and nonlinear generation (NLG) model. Table II details the quantitative results.

To study the effect of spectral mixing prior, we conduct a reference module without sum-to-one and nonnegative constraints. In spectral mixing prior, the number of endmembers is much less than the number of bands in HSIs. Hence, in the reference module, we also increase the number of hidden nodes in the autoencoder of the recognition part to go against the above role. We set it equal to the number of bands in the output HSI. For the MMI strategy, the reference method was trained without the loss term. In MMIF, the generation part aims at degeneration adaption. An inaccurate adaption may affect the final fusion performance. In the origin model, we apply the NLG model. We conduct the reference module with a linear generation model that drops all of the nonlinear activation.

As shown in the first line of Table II, the original method greatly outperforms the reference without the MMI strategy, which verifies its pivotal role. In theory, the MMI strategy promotes the inference model to match the spatial information in hr-MSI and spectral information in lr-HSI. Without the MMI strategy, the inference model estimates hr-HSI only under the constraint of the generative model and the whole algorithm turns to an inverse problem solver. The inaccurate degradation models lead to poor performance.

In the second line, we verify the effort of the unmixing mechanism. The sum-to-one and nonnegative constraints in the unmixing mechanism regularize the network away from overfitting, which results in a better performance.

In the third line of Table II, we can find that the performance gap between the linear and NLG parts in Ori CAVE is more evident than in Syn CAVE. Therefore, the linear degeneration model may hamper the performance in real scenarios.

2) Modules Analysis: We evaluated the interpretability of Am(·) and Em(·) with the visualized insight of their inner mechanisms. In Fig. 5, we show the learned abundances

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TABLE III
AVERAGE QUANTITATIVE COMPARISONS ON THE CAVE DATASET (BEST RESULTS ARE HIGHLIGHTED IN BOLD)

Methods	CNMF [11]	FUSE [13]	HySure [12]	CSTF [8]	LTTR [9]	MHFNet [10]	CUCaNet [15]	UAL[32]	Ours
Ori CAVE dataset									
PSNR	27.03	22.69	20.72	23.69	19.56	37.94	36.33	37.13	**38.04**
SAM	17.15	18.75	24.37	21.10	24.18	7.40	10.23	9.61	**5.99**
SSIM	0.84	0.68	0.60	0.63	0.57	0.96	0.92	0.94	**0.97**
ERGAS	1.88	2.25	3.37	2.10	3.39	0.47	0.71	0.78	**0.45**
Syn CAVE dataset									
PSNR	27.16	27.30	28.68	37.88	33.50	**39.01**	37.39	37.99	38.79
SAM	17.63	17.15	21.80	9.14	10.62	6.25	10.66	9.77	**5.98**
SSIM	0.89	0.84	0.88	0.95	0.95	0.97	0.97	0.96	**0.98**
ERGAS	1.73	1.47	1.76	0.46	0.79	0.44	0.65	0.73	**0.42**

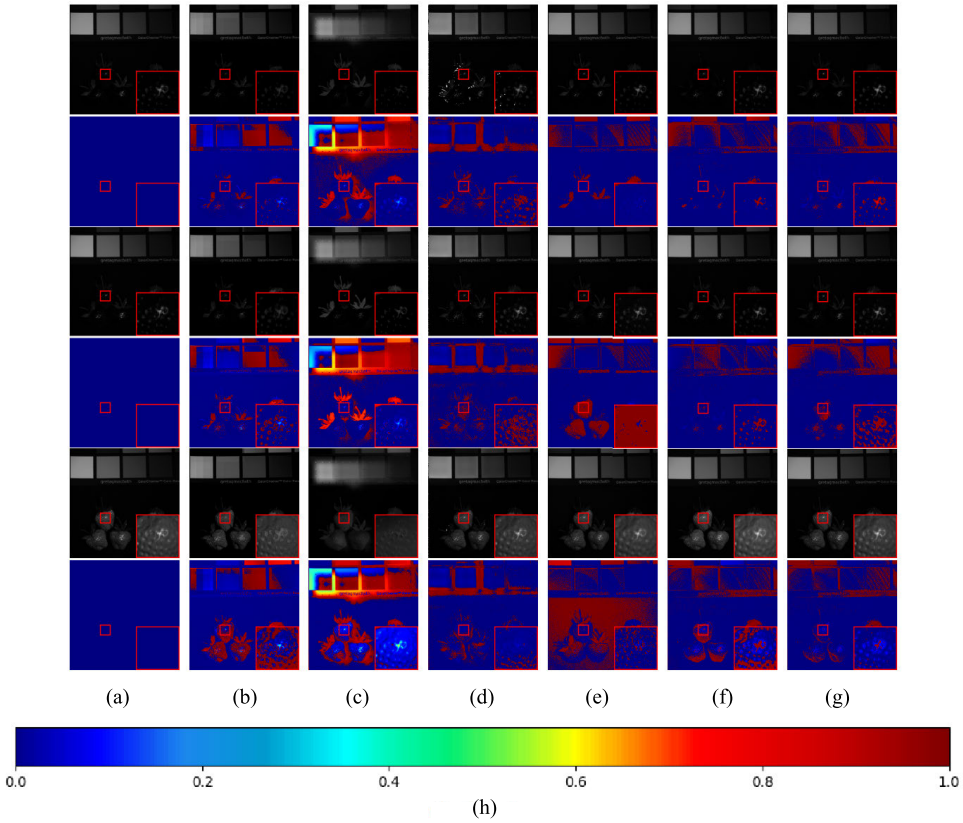


Fig. 6. Visual SR results and the corresponding error images on the scene of fake and *real* strawberries in the Ori CAVE dataset, where we display the 13th, 18th, and 22th (520, 580, 620 nm) bands of the hr-HSI images. (a) GT. (b) CNMF. (c) LTTR. (d) MHFNet. (e) CUCaNet. (f) UAL. (g) Ours. (h) Color bar.

and endmembers produced by the network. The abundances have a very clear classification of different objects, and the corresponding endmembers are also clearly distinguishable. In a real unmixing mechanism, the abundance represents the category of each pixel and the extracted endmembers are corresponding to spectral vectors of different materials. Evidently, the implemented modules can be well interpolated by the unmixing mechanism.

C. Experiments on Synthetic Data

Three HSI synthetic datasets were adopted, including Ori CAVE, Syn CAVE, and Chikusei [37].

1) Performance Comparison on CAVE: Since some of the comparative methods belong to supervised learning approaches, we selected 16 scenes from the CAVE dataset as the training set for these methods. The remaining scenes were used as the test set to compare the performance of all methods. Quantitative and visual results on the Ori CAVE dataset are shown in Table III and Fig. 6, respectively. From the visual results, we can find that the deep-learning-based methods obviously outperform traditional methods in spatial detail recovery. Visual results reveal the superiority of MMIF in global contexts and local details.

From Table III, we can find that the proposed MMIF outperforms other competing methods with respect to all metrics, especially in SAM.

The quantitative results of Syn CAVE are reported in Table III. The advantage of MMIF in Syn CAVE is less obvious than that in Ori CAVE because Syn CAVE is generated by linear operations and compared methods model degeneration in a linear way. As shown in Fig. 7, MMIF preserves a good visual impression.

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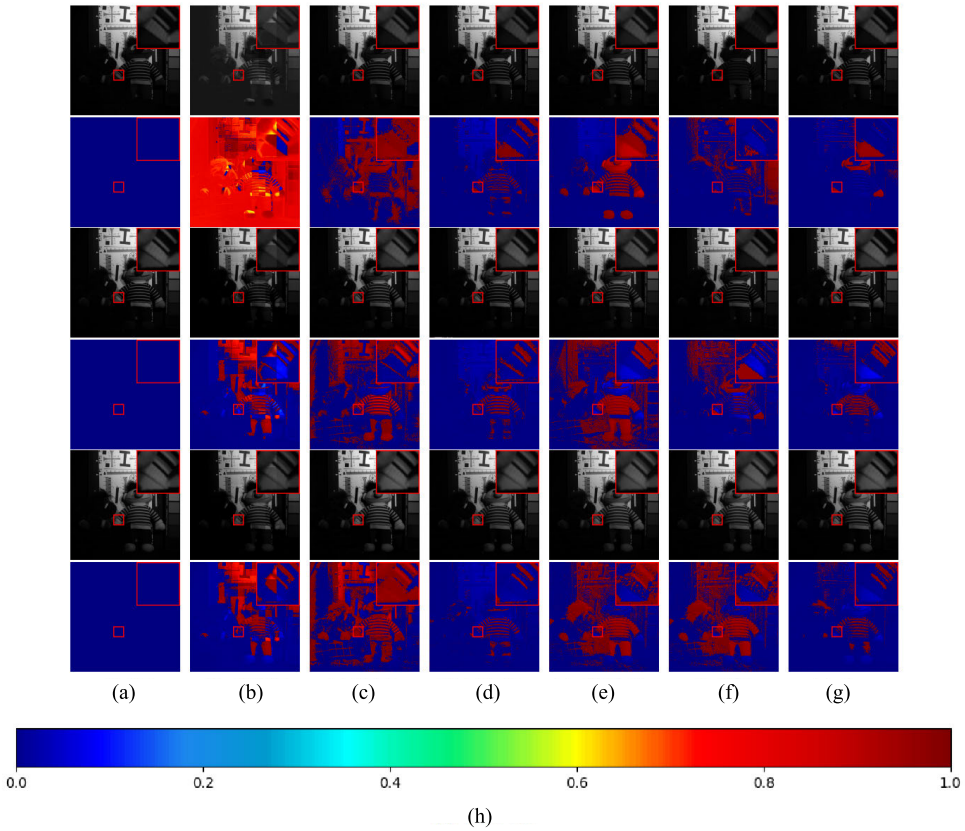


Fig. 7. Visual SR results and the corresponding error images on the scene of chart and stuffed toy in the Syn CAVE dataset, where we display the 13th, 18th, and 22th (520, 580, 620 nm) bands of the hr-HSI images. (a) GT. (b) CNMF. (c) LTTR. (d) MHFNet. (e) CUCaNet. (f) UAL. (g) Ours. (h) Color bar.

TABLE IV
AVERAGE QUANTITATIVE COMPARISONS ON THE CHIKUSEI DATASET (BEST RESULTS ARE HIGHLIGHTED IN BOLD)

Method	CNMF [11]	FUSE [13]	HySure [12]	CSTF [8]	LTTR [9]	MHFNet [10]	CUCaNet [15]	UAL[32]	Ours
PSNR	23.87	24.79	20.78	24.00	28.43	34.22	34.15	31.79	**34.23**
SAM	12.15	15.27	15.20	8.61	9.54	6.84	6.16	7.08	**5.08**
SSIM	0.76	0.50	0.63	0.95	0.69	0.83	0.82	0.67	**0.88**
ERGAS	2.89	1.53	2.57	1.67	1.43	0.67	0.63	0.93	**0.58**

2) Performance Comparison on Chikusei: Chikusei is an airborne hyperspectral dataset with 2517 × 2335 pixels and 128 bands in the spectral range from 363 to 1018 nm. We generate the corresponding hr-MSI by employing the SRF of the Landsat sensor. We set the scale factor as 32 and simulated the lr-HSI under Gaussian blur kernel with the same size as the scale factor and deviation of 0.5. We select the right bottom of Chikusei with 2048 × 2048 pixels and crop it to 16 nonoverlapped images with the size of 512 × 512 for the experiment. Among them, nine images were employed for testing, and the left images were used for training.

Table IV shows the average performance of all competing methods. It can be seen that MMIF outperforms all competing methods in all metrics, especially SAM. Benefitting from the MMI strategy, MMIF has a strong ability to extract spectral information, so it has a great advantage in SAM. Fig. 8 reports the visual results. MMIF takes full advantage of detail reconstruction. The recovered spectral vectors of different methods are visualized in Fig. 9. The proposed MMIF can grasp the spectral characteristics more accurately.

TABLE V
AVERAGE PERFORMANCE ON THE SYN CAVE DATASET WITH DIFFERENT WEIGHTS OF THE MMI STRATEGY LOSS TERM β

β	0.1	0.2	0.3	0.4	0.5	0.6
PSNR	38.11	38.27	38.56	38.71	**38.79**	38.79
SAM	6.34	6.28	6.11	6.04	**5.98**	5.99
SSIM	0.94	0.96	0.97	**0.98**	**0.98**	**0.98**
ERGAS	0.47	0.46	0.44	0.43	**0.42**	**0.42**

TABLE VI
MODEL SIZE OF DEEP-LEARNING-BASED METHODS

Methods	MHFNet	CUCaNet	UAL	Ours
Model Size (M)	129.4	2.28	27.05	0.021

D. Experiments on Real Data

We adopt a real dataset, Worldview2, in this part. Worldview2 contains an lr-HSI of size 419 × 658 × 8 and an

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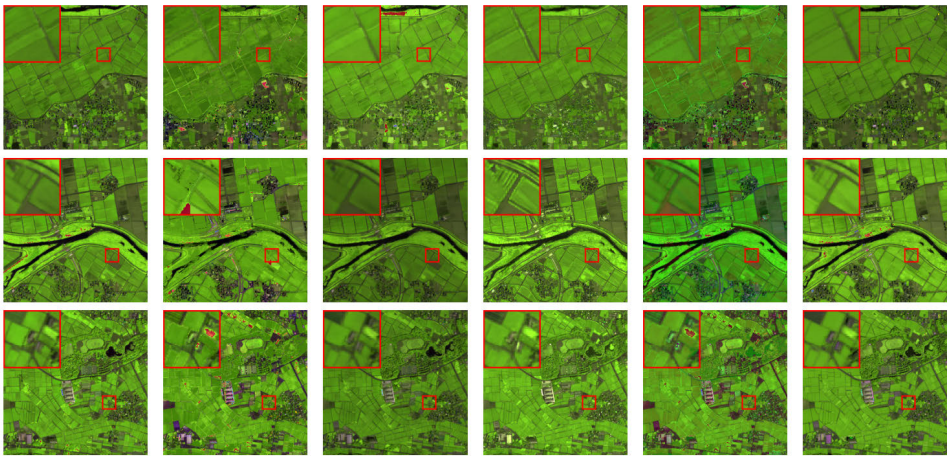


Fig. 8. Visual SR results in the Chikusei dataset. We display three test images with bands 70-100-36 as R-G-B to show. (a) GT. (b) CNMF. (c) MHFNet. (d) CUCaNet. (e) UAL. (f) Ours.

TABLE VII

AVERAGE HSI-SR RESULTS OF THE TEST METHODS ON THE SYN CAVE DATASET GENERATED WITH DIFFERENT BLUR KERNELS WITH σ OUT OF RANGE [0.5, 5.0]

sigma	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
MMIF(Ours)										
PSNR	38.79	38.76	38.78	38.75	38.76	38.78	38.78	38.77	38.78	38.77
SAM	5.98	6.04	6.01	6.02	6.01	6.00	6.02	6.00	6.00	6.03
SSIM	0.98	0.97	0.98	0.97	0.98	0.98	0.98	0.98	0.98	0.97
ERGAS	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42
CUCaNet [15]										
PSNR	37.39	37.31	37.32	37.35	37.33	37.35	37.35	37.31	37.36	37.34
SAM	10.66	10.98	10.96	10.91	10.92	10.92	10.94	10.95	10.94	10.93
SSIM	0.97	0.93	0.92	0.94	0.94	0.93	0.93	0.94	0.94	0.94
ERGAS	0.65	0.68	0.67	0.68	0.68	0.68	0.67	0.68	0.69	0.68

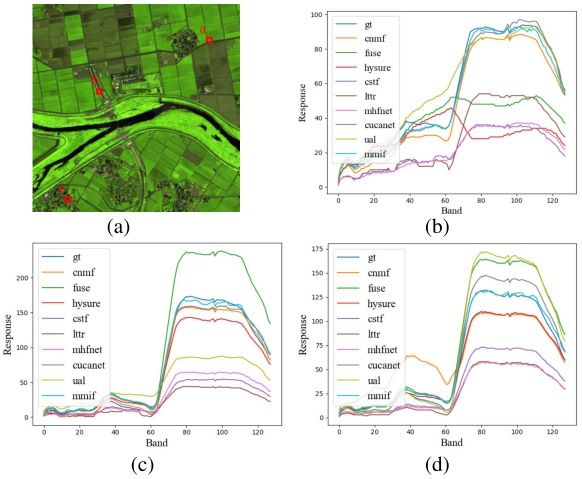


Fig. 9. Visualization of spectral vectors on image 2 of the chikusei dataset. (a) RGB image. (b) Spectral vectors on point a. (c) Spectral vectors on point b. (d) Spectral vectors on point c.

hr-MSI of size $1676 \times 2632 \times 3$, while the hr-HSI is unavailable.

In the experiment, the left top of lr-HSI ($384 \times 640 \times 8$) and hr-MSI ($1536 \times 2560 \times 3$) were cropped to 15 image pairs. To train the competing supervised method, we build a training set with the Wald protocol [38], where both lr-HSI and hr-MSI are downsampled spatially to build an observation image, and the original lr-HSI is used as ground-truth hr-HSI in the training set. In Worldview2, the true hr-HSI is unavailable. Hence, we only have visual results. As shown in Fig. 10, the reconstruction of MMIF holds clearer edges to hr-MSI.

E. Parameter Analysis

In this part, we analyze the influence of different parameter settings on performance. The index results under different weights of MMI strategy loss are shown in Table V. Within a certain range, as the weight increases, the indices of the proposed method are better, which indicates the significant effort of the MMI strategy. Moreover, we analyze the relationship between MMI strategy loss and performance in each training epoch. As shown in Fig. 11, the metric is inversely proportional to the loss item, which confirms the positive effect of the MMI strategy on the performance.

In the above experimental studies, the blur kernel B is set as a Gaussian kernel with a standard deviation of 0.5. However, the degradation parameter varies in the real scenario. We build a set of CAVE datasets with different settings of blur kernel for generating lr-HSI. The performance presented in

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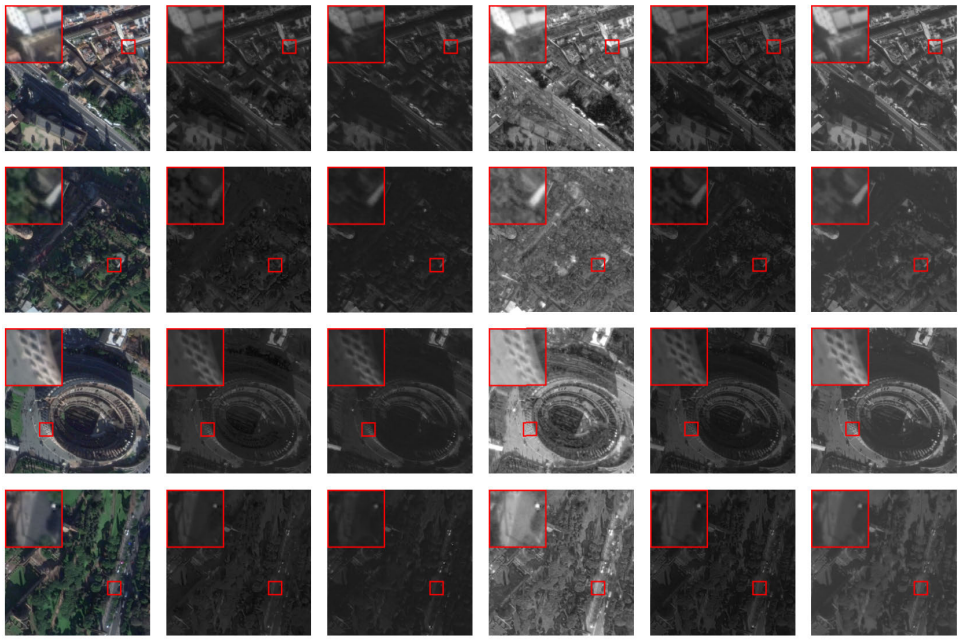


Fig. 10. Visual SR results in the WV2 dataset. We show the first band of the reconstructed images and the corresponding MSIs. (a) hr-MSI. (b) CNMF. (c) MHFNet. (d) CUCaNet. (e) UAL. (f) Ours.

TABLE VIII
AVERAGE RUNTIME ON THE SYN CAVE DATASET WHEN SR SCALE IS 32

Methods	CNMF	FUSE	HySure	CSTF	LTTR	MHFNet	CUCaNet	UAL	Ours
Training Time (min)	0	0	0	0	0	1255	0	351	0
Testing Time (s)	101.2	4.01	153.8	123.8	327.6	23.9	2052.7	29.7	1393.2

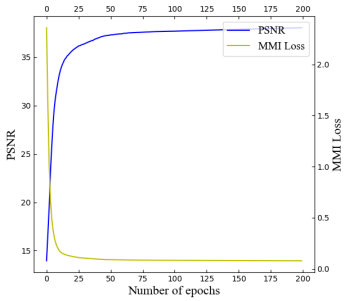


Fig. 11. Trends of PSNR and MMI loss during training. The performance index is inversely proportional to the MMI loss term, which indicates the positive effort of the MMI strategy on the reconstruction.

Table VII indicates the great robustness of MMIF on different degradation parameters.

F. Runtime Analysis

We evaluate the average runtime of all methods on each image in the CAVE dataset when the scale factor is 32. All traditional methods run on a desktop with AMD Ryzen 5 3600X 6-Core Processor at 3.79 GHz. All deep-learning-based methods run on an NVIDIA RTX 3090 GPU with 24GB memory. As shown in Table VIII, the runtime of MMIF is acceptable and shorter than most unsupervised DL-based compared methods.

As shown in Table VI, our method surpasses other DL-based methods by a large margin in lightweight.

VI. CONCLUSION

In this article, we have proposed a new unsupervised HSI-SR algorithm with efficient information fusion and presented a deep-learning-based implementation for the algorithm without harming the interpretability. The proposed MMIF handles the HSI-SR problem from three factors, that is, domain knowledge of HSIs, reliable information interaction between observed images, and interpretable data representation. We first build a new VI architecture for HSI-SR, simulating the generation and degradation in HSIs. For an efficient information fusion, the MMI strategy is proposed for information interaction between lr-HSI and hr-MSI with a quantitative index. Combining the MMI strategy with the VI architecture for an end-to-end algorithm is straightforward. Finally, we show how to implement the algorithm with a neural network without losing interpretability. Comprehensive experiments implemented on synthetic and real datasets verify the superiority of MMIF and substantiate the rationality of the proposed MMI strategy.

There are two directions we leave in our future work. Recent deep-learning-based unsupervised HSI-SR methods optimize on one image pair with many iterations on training, which makes it challenging to deploy in real-time applications. More seriously, in some application scenarios, the deployed device

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cannot support the training of unsupervised deep-learning methods. The first future research line is to explore a fast unsupervised deep-learning-based HSI-SR method with one-step inference as supervised methods. In the MMIF, we utilize an unmixing mechanism at the VI architecture. In essence, the unmixing mechanism is a classifier that predicts the category of each pixel. Another future work is to realize HSI-SR and classification simultaneously.

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